

Beyond Telescoping

Weighted Zero-Sum Invariants in Hexation and Multi-Asset Financial Networks

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Abstract

Ghosh's Hexation paper defines six positions from three assets - a bond, a stock, and gold - and records both ordinary and asset-value-weighted identities that sum to zero [1]. This paper generalizes those identities through graph incidence matrices, cycle spaces, and potential differences. For asset values assigned to the vertices of a directed network, positions are differences along edges. Every closed cycle then yields a zero-sum identity; every scalar potential generates a corresponding weighted family. Polynomial potentials produce an infinite hierarchy of explicit weights, while complete networks also admit reciprocal and triangular identities. The article distinguishes algebraic invariants from independent constraints and from economically implementable trading strategies. It outlines an empirical evaluation programme and develops carefully delimited connections to finance, econometrics, physics, chemistry, biology, psychology, philosophy, political science, diplomacy, geopolitics, welfare economics, and computational systems. The mathematical results are exact; the financial usefulness, stability, and welfare effects of any implementation remain empirical questions.

The paper ends with "The End"

1 Introduction

The central question is not merely whether positions telescope to zero, but which alternative weighted sums also vanish, how those identities depend on network structure, and which are independent. The starting point is the author's *Hexation*, whose standard configuration uses a bond value $B(t)$, stock value $S(t)$, and gold value $G(t)$ to define six directed differences. The manuscript interprets these positions as a prolonged economic choice involving participating firms, financial institutions, and governance [1].

The present paper supplies a graph-theoretic and potential-based classification. Incidence matrices and cycle spaces provide a standard language for networks [2, 3, 4]. The main warning is equally important: a zero identity among quoted asset values does not by itself imply an arbitrage, a positive expected return, risk reduction, or social benefit. Financial outcomes depend on quantities, contractual terms, transaction costs, liquidity, financing, counterparty exposures, and the way positions are actually implemented [5, 6].

2 Standard Hexation and Its Identities

Let the asset values at time t be $B(t)$, $S(t)$, and $G(t)$. Standard Hexation defines

$$a(t) = B(t) - G(t), \quad a'(t) = G(t) - B(t), \quad (1)$$

$$b(t) = S(t) - B(t), \quad b'(t) = B(t) - S(t), \quad (2)$$

$$c(t) = G(t) - S(t), \quad c'(t) = S(t) - G(t). \quad (3)$$

The first pair of zero-sum identities is

$$a(t) + b(t) + c(t) = 0, \quad (4)$$

$$a'(t) + b'(t) + c'(t) = 0. \quad (5)$$

These are telescoping sums. Hexation also gives the asset-value-weighted identities

$$S(t)a(t) + G(t)b(t) + B(t)c(t) = 0, \quad (6)$$

$$S(t)a'(t) + G(t)b'(t) + B(t)c'(t) = 0. \quad (7)$$

For example, the first weighted identity follows directly from

$$S(B - G) + G(S - B) + B(G - S) \quad (8)$$

$$= SB - SG + GS - GB + BG - BS = 0. \quad (9)$$

These relations are algebraic identities given the stated definitions; their presence does not, without further assumptions, establish the profitability or stability of the corresponding positions [1].

3 Asset Networks and Cycle Invariants

Let $\mathcal{G} = (V, E)$ be a finite directed graph with n vertices and m edges. Each vertex i represents an asset with comparable value A_i . For a directed edge $e = (i, j)$, define

$$x_{ij} = A_i - A_j. \quad (10)$$

Let D denote the vertex-edge incidence matrix, with $+1$ at the tail of each directed edge and -1 at its head. If A is the vector of asset values, the edge-position vector is

$$x = D^T A. \quad (11)$$

A cycle vector z assigns oriented unit flow to a closed cycle. It satisfies $Dz = 0$. Therefore

$$\boxed{z^T x = z^T D^T A = (Dz)^T A = 0.} \quad (12)$$

This is the graph form of telescoping: every closed cycle of potential differences sums to zero. For a graph with c connected components, its cycle space has dimension $m - n + c$; for a connected graph this becomes $m - n + 1$ [2, 3, 4]. This dimension counts independent cycle relations in the edge space, not the number of all possible written identities.

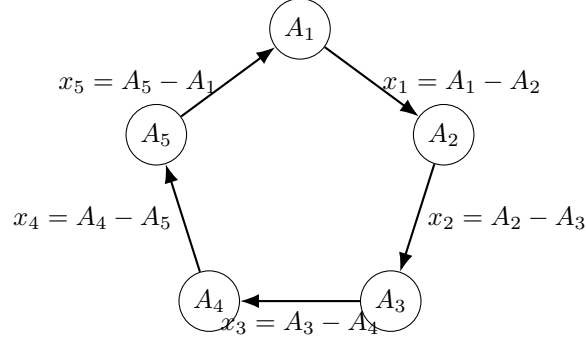


Figure 1: A five-asset cycle. The directed differences telescope around the closed loop.

For the five-asset cycle in Figure 1,

$$\boxed{\sum_{i=1}^5 x_i = 0.} \quad (13)$$

More generally, the identity holds for any closed cycle, irrespective of whether the network contains every possible pair of assets.

4 Potential-Weighted Zero-Sum Families

The central generalization follows by replacing the identity potential with a scalar function. Let ϕ be a function on the set of asset values. For edge (i, j) with $A_i \neq A_j$, define the divided-difference weight

$$w_{ij}^{(\phi)} = \frac{\phi(A_i) - \phi(A_j)}{A_i - A_j}. \quad (14)$$

If ϕ is differentiable, the continuous extension at $A_i = A_j$ is $\phi'(A_i)$. By definition,

$$w_{ij}^{(\phi)} x_{ij} = \phi(A_i) - \phi(A_j). \quad (15)$$

Theorem 1 (Potential-weighted cycle identity). For any closed directed cycle C and any scalar potential ϕ for which the weights are defined,

$$\boxed{\sum_{(i,j) \in C} w_{ij}^{(\phi)} x_{ij} = 0.} \quad (16)$$

Proof. Substitution gives the sum of successive potential differences $\phi(A_i) - \phi(A_j)$. Every intermediate term cancels around the closed cycle, leaving zero. $\square$

Choosing polynomial potentials $\phi(u) = u^p$, for integer $p \geq 1$, gives

$$w_{ij}^{(p)} = \sum_{r=0}^{p-1} A_i^{p-1-r} A_j^r, \quad (17)$$

so

$$\boxed{\sum_{(i,j) \in C} \left(\sum_{r=0}^{p-1} A_i^{p-1-r} A_j^r \right) x_{ij} = 0.} \quad (18)$$

The first three members are

$$p = 1 : \sum_{(i,j) \in C} x_{ij} = 0, \quad (19)$$

$$p = 2 : \sum_{(i,j) \in C} (A_i + A_j)x_{ij} = 0, \quad (20)$$

$$p = 3 : \sum_{(i,j) \in C} (A_i^2 + A_i A_j + A_j^2)x_{ij} = 0. \quad (21)$$

Thus the ordinary telescopic identity is one member of an infinite family of potential-weighted identities. The family is mathematically exact; it does not imply that all its members are linearly independent or economically distinct.

5 Reciprocal and Triangular Identities

For any pair of assets,

$$x_{ij} + x_{ji} = 0. \quad (22)$$

For any triple (i, j, k) , the triangular cycle identity is

$$x_{ij} + x_{jk} + x_{ki} = 0. \quad (23)$$

A value-dependent weighted triangle identity also holds:

$$\boxed{A_i x_{jk} + A_j x_{ki} + A_k x_{ij} = 0.} \quad (24)$$

To verify it, expand the expression as

$$A_i(A_j - A_k) + A_j(A_k - A_i) + A_k(A_i - A_j) \quad (25)$$

$$= A_i A_j - A_i A_k + A_j A_k - A_i A_j \quad (26)$$

$$+ A_i A_k - A_j A_k = 0. \quad (27)$$

This last identity is nonlinear in the asset values because its coefficients depend on those same values. It should not be counted automatically as an independent linear constraint on the edge-position vector.

Table 1: Families of zero-sum identities and their interpretation.

Family	Representative identity	Structural meaning
Cycle closure	$\sum_C x_{ij} = 0$	Potential differences close around a cycle
Polynomial potential	$\sum_C (A_i + A_j)x_{ij} = 0$	Difference of squared values telescopes
Higher potential	$\sum_C w_{ij}^{(p)} x_{ij} = 0$	Difference of a polynomial potential telescopes
Reciprocity	$x_{ij} + x_{ji} = 0$	Reversing an oriented difference changes its sign
Weighted triangle	$A_i x_{jk} + A_j x_{ki} + A_k x_{ij} = 0$	Value-dependent algebraic cancellation

6 Independence, Identification, and the Research Question

The existence of many equations does not imply the existence of equally many independent restrictions. The linear cycle identities are organized by the cycle space. Some written identities are combinations of others, and potential-weighted identities introduce coefficients that depend on asset values. A rigorous classification should therefore distinguish at least three concepts: (i) an algebraic identity, (ii) an independent restriction under a specified model, and (iii) a relation that has an economically interpretable mechanism.

A convenient computational test is to construct the incidence matrix, compute its rank, and use a basis for its null space to enumerate independent cycle identities. For nonlinear potential-weighted identities, symbolic algebra can verify identities and numerical rank calculations can explore local independence over specified parameter regions. Numerical rank is evidence for a chosen sample and tolerance, not a universal proof of functional independence.

7 Financial Interpretation and Limits

For an economic application, A_i must be defined carefully. Prices should be expressed in a common numeraire and observation time, with units, contract multipliers, coupon and dividend cash flows, maturity, and any conversion rules handled consistently. A quoted bond price, stock price, and gold price are not automatically comparable economic quantities without such conventions.

The algebra concerns differences of values. An implementable strategy requires positions in quantities or contracts, financing, rebalancing rules, counterparties, and a loss function. Transaction costs and liquidity affect the economics of trade [5, 7]. Interconnected financial markets can propagate stress through liquidity hoarding, funding links, and fire-sale externalities, so network complexity must be accompanied by risk controls rather than treated as a source of profit by itself [6].

In particular,

$$\text{zero-sum identity} \not\Rightarrow \text{arbitrage}, \quad (28)$$

and

$$\text{algebraic closure} \not\Rightarrow \text{positive expected return}. \quad (29)$$

A useful research programme asks whether the invariant structures improve hedging, attribution, capital allocation, robustness, or monitoring compared with simpler baselines. The answer depends on the empirical model and the market environment.

8 Econometrics and Statistics

The identities are deterministic conditional on correctly measured asset values and consistent definitions. Therefore, a test that reconstructs positions from the same A_i values primarily checks data integrity, edge orientation, units, and implementation. It does not independently validate a claim about profits or stability.

Let Y_{ft} measure a prespecified financial or risk outcome for network configuration f at time t . A descriptive panel specification might be

$$Y_{ft} = \alpha_f + \tau_t + \beta Z_{ft} + \gamma^\top X_{ft} + \varepsilon_{ft}, \quad (30)$$

where Z_{ft} indicates use of a specified invariant-based allocation rule, X_{ft} contains controls, and α_f and τ_t represent unit and time effects. Causal interpretation requires a defensible identification strategy; the equation alone is not causal evidence [8, 9].

An evaluation should pre-register outcome definitions and compare out-of-sample performance against baselines while accounting for turnover, transaction costs, leverage, liquidity, tail

losses, and model-selection effects. Statistical uncertainty should be reported, and a successful identity check should not be mistaken for a successful economic backtest.

9 Algorithms, Data Science, and AI/ML

A reproducible workflow can be implemented without treating every identity as a separate strategy.

Algorithm: EnumerateCycleInvariants

Input:

comparable asset values A
directed asset network $G=(V,E)$
candidate potential functions Φ

1. Validate dates, units, numeraire, and missing values.
2. Build the incidence matrix D with declared edge orientation.
3. Compute edge positions $x = \text{transpose}(D) * A$.
4. Compute a basis for the cycle space $\text{null}(D)$.
5. Verify $z' * x = 0$ for each basis cycle z .
6. For each ϕ in Φ :
 - construct divided-difference edge weights;
 - evaluate weighted cycle identities;
 - record numerical residuals and tolerances.
7. Label identities as algebraic, independent under a model, or economically hypothesized; do not conflate the labels.
8. If evaluating a trading application, apply costs, risk limits, out-of-sample tests, and benchmark comparisons separately.

Output:

identity catalogue, rank diagnostics, residual report,
provenance record, and (if tested) economic evaluation.

AI/ML can assist with anomaly detection, symbolic-conjecture generation, parameter estimation, clustering networks, and risk forecasting. It should not replace algebraic proof with pattern matching. A learned model may propose a weighting function, but symbolic verification and out-of-sample economic validation remain separate gates [10, 11].

10 Connections to Other Fields

The same formal structure has useful analogies across disciplines, provided that an analogy is not confused with a claim of shared causal mechanism.

Table 2: Interdisciplinary connections and their limits.

Field	Connection	Limit or practical question
Physics	A potential difference around a closed loop sums to zero in a conservative field.	Financial positions are not physical fields; the analogy is mathematical.
Chemistry	Reaction-network cycles can be represented using stoichiometric and incidence matrices.	A cycle identity does not establish chemical equilibrium or reaction feasibility.
Biology	Network feedback and cycles organize interactions in systems biology.	Biological stability requires dynamical assumptions beyond graph closure [13].
Psychology	Framing and loss aversion may affect how agents perceive the same exposure.	Perception is distinct from objective return and risk [12].
Psychiatry	Stress and bounded attention are relevant to institutional decision processes.	No clinical diagnosis follows from a financial-network model.
Philosophy	A formal representation is distinct from the economic object it represents.	Definitions and assumptions must remain explicit.
Religion and moral economy	Stewardship and responsibility frame the governance of pooled resources.	Normative claims require explicit ethical commitments.
Political science	Governance allocates authority, responsibility, and oversight.	Institutional design must be evaluated in context.
Diplomacy and international relations	Cross-border financial links can create bargaining dependencies.	Dependence may facilitate cooperation or transmit shocks [14].
Geopolitics	Sanctions, jurisdiction, currencies, and sovereign risk change network exposures.	Country and legal constraints must be modelled explicitly.
Welfare economics	Private returns can differ from social welfare because of externalities.	Social costs and distributional effects require measurement [15].
Warfare economics	Conflict shocks can affect sovereign risk, liquidity, and supply networks.	Analysis here is limited to high-level resilience and risk assessment.
Computer science	Incidence matrices and graph algorithms yield reproducible catalogues.	Data conventions and software tests remain necessary.

The general lesson is methodological: define the state variables, represent the relationships, prove the algebraic claims, and then investigate the dynamics and welfare consequences with appropriate evidence. In financial networks, stability and efficiency can trade off, and interconnectedness can transmit stress [6]. For this reason, any proposed Hexation-based institution should be evaluated for both private incentives and system-wide externalities.

11 Governance, Welfare, and Strategic Risk

If banks finance participants or share profits, the arrangement introduces principal-agent problems, credit exposure, and potential incentives to increase risk. A governance specification should define licensing, collateral, loss allocation, disclosure, audit rights, capital requirements, conflicts of interest, and exit conditions. The original Hexation manuscript already treats bank licensing and asset regulation as governance variables [1].

A stylized social objective is

$$\max_{\pi} \mathbb{E}[R(\pi)] - \lambda \mathcal{R}(\pi) - C(\pi), \quad (31)$$

where π is a policy, R is measured benefit, $\mathcal{R}$ is a risk functional, and C includes funding, transaction, and institutional costs. The risk functional should include tail risk and interdependence rather than only average volatility. Welfare analysis should also consider who receives gains, who bears losses, and whether public backstops create moral hazard.

In geopolitical and conflict scenarios, asset networks may be affected by sanctions, payment restrictions, capital controls, or disruptions. The appropriate use of the framework is to test resilience, concentration, and risk propagation at a high level, not to infer operational outcomes from a zero-sum identity. Diplomatic communication, legal compliance, and transparent governance are therefore part of the institutional setting, not optional decorations.

12 Limits and Research Agenda

The mathematical results in this paper follow from definitions and standard graph theory. They establish families of zero-sum identities, not a new law of market prices. Several questions remain open for the Hexation research programme:

1. Which potential-weighted identities are distinct after accounting for algebraic dependence?
2. Which network topologies provide a transparent and economically interpretable basis of positions?
3. Under what contract, numeraire, and quantity conventions can the positions be implemented consistently?
4. Do any identity-based allocation rules improve risk-adjusted, after-cost, out-of-sample outcomes?
5. What governance mechanisms prevent leverage, liquidity, counterparty, and systemic risks from dominating potential benefits?

These questions require a combination of symbolic mathematics, graph computation, financial modelling, econometric identification, and institutional evaluation. A negative empirical result would be informative: algebraic elegance alone is not a sufficient economic criterion.

13 Conclusion

Hexation's simple telescoping sum is the first member of a much broader family. Representing asset values as node potentials and positions as edge differences yields cycle identities; applying scalar potentials produces weighted identities of arbitrary polynomial degree. Reciprocal and triangular constructions give additional relations, while graph theory clarifies which linear cycle constraints are independent.

The central chain is

$$\boxed{\text{Assets} \rightarrow \text{Network} \rightarrow \text{Positions} \rightarrow \text{Invariants} \rightarrow \text{Economic evaluation.}} \quad (32)$$

The transition from invariants to economic evaluation is the critical one. It requires comparable data, implementable contracts, explicit costs, robust risk management, credible identification, and attention to welfare and governance. The paper’s central result is therefore both mathematical and methodological: discover the zero sums, classify their independence, and test their economic significance without confusing an exact identity with a profitable strategy.

References

- [1] S. Ghosh, *Hexation*, Kolkata, India.
- [2] F. Harary, *Graph Theory*, Addison-Wesley, Reading, Massachusetts, 1969.
- [3] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, Upper Saddle River, New Jersey, 2001.
- [4] G. Strang, *Graphs, Networks, Incidence Matrices*, MIT OpenCourseWare, Linear Algebra, 2011. <https://ocw.mit.edu/courses/18-06sc-linear-algebra-fall-2011/>.
- [5] Y. Amihud and H. Mendelson, Asset Pricing and the Bid-Ask Spread, *Journal of Financial Economics*, 17(2), 223–249, 1986.
- [6] I. Aldasoro, D. Delli Gatti, and E. Faia, Bank Networks: Contagion, Systemic Risk and Prudential Policy, *BIS Working Papers*, No. 597, Bank for International Settlements, 2016. BIS Working Paper 597 webpage.
- [7] Bank for International Settlements, The Economics of Market-Making, *BIS Quarterly Review*, March 2015. <https://www.bis.org/publications/economics-market-making>.
- [8] J. M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, Cambridge, Massachusetts, 2010.
- [9] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, Cambridge, 2009.
- [10] C. E. Shannon, A Mathematical Theory of Communication, *Bell System Technical Journal*, 27, 379–423 and 623–656, 1948.
- [11] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, Cambridge, Massachusetts, 2018.
- [12] D. Kahneman and A. Tversky, Prospect Theory: An Analysis of Decision under Risk, *Econometrica*, 47(2), 263–291, 1979.
- [13] U. Alon, *An Introduction to Systems Biology: Design Principles of Biological Circuits*, Chapman and Hall/CRC, Boca Raton, Florida, 2006.
- [14] R. O. Keohane and J. S. Nye, *Power and Interdependence: World Politics in Transition*, Little, Brown, Boston, 1977.
- [15] K. J. Arrow, *Social Choice and Individual Values*, 2nd ed., Yale University Press, New Haven, Connecticut, 1963.

Glossary

Asset value

A price or value measure attached to an asset at a specified time, in a specified numeraire and unit.

Position

In this paper, a directed difference $x_{ij} = A_i - A_j$; an implementable financial position additionally requires quantities and contractual terms.

Incidence matrix

A matrix encoding the start and end vertices of the directed edges of a graph.

Cycle space

The null space of the incidence matrix; its elements encode circulations around closed network cycles.

Potential

A scalar function assigned to vertices or asset values, such as $\phi(A_i) = A_i^p$.

Divided difference

The ratio $(\phi(A_i) - \phi(A_j))/(A_i - A_j)$, with a derivative extension where appropriate.

Zero-sum invariant

An identity in which a specified sum of positions or weighted positions equals zero under the model definitions.

Independent constraint

A restriction not implied by the other specified restrictions within the stated model.

Numeraire

The unit in which asset values and cash flows are expressed.

Liquidity risk

The risk that an asset or position cannot be traded or funded at acceptable cost when needed.

Systemic risk

Risk transmitted through interactions and dependencies among institutions or markets.

Out-of-sample validation

Evaluation on data not used to fit or select the model.

The End